

1. DEFINITION

DEFINITION
Propylene glycol monolaurate (type II) EP/NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Propylene glycol laurate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	668Z5835Z3 (Propylene glycol monolaurate)
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Propylene glycol and Lauric acid. The esterification step is followed by a distillation in order to obtain a high level of monoesters. No solvent or catalyst used during the manufacturing process.

COMPOSITION	
Well defined multi-constituent substance composed of Propylene glycol mono- and di- esters of Lauric acid (C12), the monoester fraction being predominant (Specification > 90%)	
Full Labeling (Key letters system) $A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
Propylene glycol laurate	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	



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4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	7818
Canada MF (Type III – Excipient)	/
For submission of the DMF in another country, please contact your local Gattefossé representative	

CHINA BUNDLING REVIEW			
(Generic name) Chinese name	Registration N°	Company	CFDA/CDE Reference (http://www.cde.org.cn/yfb.do?method=main)
Propylene Glycol Monolaurate Type II 单月桂酸丙二醇酯 (II型)	F20220000058	Gattefossé SAS	Raw material has not been approved yet for use in marketed formulations in Joint-Review with drug applications (Status I)
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopeia Current edition	Propylene glycol monolaurate (type II)
USP/ NF Current edition	Propylene glycol monolaurate (type II)

6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓ (9) warning: cat	✓ (1) All species (No limit)	✓ (9) warning: cat	✓ (5) USFA 172.856	✓ (5) 40CFR180.1250	✓ (5) 40CFR180.1250



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EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
EUROPEAN REGULATION (1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption) (2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units) (3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether) (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin USA REGULATION - Veterinary drugs FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR) (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database) - Pesticides (8) EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)					

7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Propylene glycol monolaurate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicine	/
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	Propylene glycol laurate	Role Non-medicinal: Purposes: Emulsifying Agent , Skin-Conditioning Agent - Emollient Acceptable Daily Intake Upper Limit: 25.0 Milligrams/kilogram body weight/day Route Of Administration: Oral Allowed Purposes: Emulsifying Agent General: Route Of Administration: Topical Allowed Purposes: Skin-Conditioning Agent - Emollient Detail: Topical use only.



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REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA [NHPID](#)

Status	NHPID name	Route of administration/ Restrictions
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute irritation (OECD 404)	Dermal	Rabbit	LAUROGLYCOL™ 90 pure (undiluted)	Moderately Irritant	(Gattefossé data, GAT -99441, 1999)
Acute irritation (OECD 405)	Ocular	Rabbit	LAUROGLYCOL™ 90 pure (undiluted)	Slightly Irritant	(Gattefossé data, GAT -99440, 1999)
Acute toxicity (OECD 401)	Oral	Rat	LAUROGLYCOL™ 90 pure (undiluted)	LD50(oral) > 2003 mg/kg	(Gattefossé data, GAT -99444, 1999)
Acute toxicity (Dose escalating)	IV bolus (tail vein)	Mouse	LAUROGLYCOL 90 Vehicle: Physiological saline solution M: 25, 50, 100, 200 and 150 mg/kg F: 25, 50, 100, 200, 400, 300, 250 and 200 mg/kg	MTD(IV) (M): 150 mg/kg MTD(IV) (F): 200 mg/kg	(Gattefossé SAS data, GAT-MTDs-PH- 08/0476, 2009)
Acute toxicity (eq OECD 423)	IV bolus (tail vein)	Mouse	LAUROGLYCOL 90 Vehicle: Physiological saline solution M+F: 150 mg/kg	MTD(IV): 150 mg/kg M+F: Administration at 150 mg/kg induces no mortality and toxicity signs.	(Gattefossé SAS data, GAT-TAIs-PH- 08/0476, 2009)
Ames test (With and without activation) (OECD 471)	In vitro	S. typhimurium	LAUROGLYCOL™ 90 pure	Negative. Not mutagenic.	(Gattefossé data, GAT-99462, 1999)

CIR COMPENDIUM (Cosmetic Ingredient Review) (<http://www.cir-safety.org>)

Denomination	Reference	Maximum “as used” concentration	Conclusions
Propylene glycol mono and diesters	IJT, 18 (Suppl. 2) 35-52, 1999	Up to 50 %	Based on the available animal and clinical data included in this report, the CIR Expert Panel concludes that Propylene Glycol laurate is as cosmetic ingredients in the present practices of use.

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EMA-Recommended Safety Limits (Excipients in the label and package leaflet of medicinal products for human use. EU Medicines Agency, CPMP/463/00 Rev. 1; publication planned end 2017)			
Propylene glycol esters	Neonates up to 28 days (or 44 weeks post menstrual age for pre-terms)	1 month (29 days) up to 4 years	5 years up to 17 years and adults
Safety limits	1 mg/kg	50 mg/kg	500 mg/kg

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	CAS #	EINECS #	Chemical name
Propylene glycol laurate	27194-74-7 (or 142-55-2)	248-315-4 (or 205-542-3)	Dodecanoic acid, monoester with 1,2-propanediol (or 2-hydroxypropyl laurate)

NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
Propylene glycol laurate	Listed	DSL listed	Listed	Listed	Dodecanoic acid, monoester with 1,2-propanediol: TSCA, DSL, ENCS No. 2-772, ECL Serial No. KE-12880, AICS, PICCS, ASIA-PAC, NZIoC, IECSC, TCSI (or 2-hydroxypropyl laurate: EINECS, ECL Serial No. KE-20875, AICS, ASIA-PAC, NZIoC, IECSC, TCSI)

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
Propylene glycol laurate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.



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10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
Propylene glycol monolaurate (UNII: 668Z5835Z3)	ORAL	Capsule		2115 mg
	ORAL	Tablet		20 mg
	ORAL	Tablet, film coated		20 mg
	TOPICAL	Gel	3% w/w	
	TRANSDERMAL	Film, extended release	NA	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.2. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENTS (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	Functional Category	Regulatory status	Safety
Propylene glycol monolaurate	Emollient Emulsifying agent Nonionic surfactant Oleaginous vehicle Penetration enhancer Solubilizing agent	Included in the FDA inactive Ingredients Database. Included in nonparenteral medicines licesed in the UK. Included in the Natural Health Products Ingredients Database.	Propylene glycol monolaurate is generally regarded as safe for use in pharmaceutical and cosmetic formulations in the present practices of use. Prolonged or frequently repeated skin contact may cause allergic reactions in hypersensitive individuals exposed to this product, and it may also cause eye irritation. LD50 (rat, oral) : 2.0 g/kg for lauroglycol 90 and lauroglycol FCC.

10.2. Food/Nutraceutical/Dietary supplement applications

Propylene glycol esters of Fatty acids have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.



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Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS 477 Propane-1,2-diol esters of fatty acids
E.U.	EFA (European Food additives)	E 477 Propane-1,2-diol esters of fatty acids
U.S.A.	DSHEA (Dietary Supplement Health and Education Act) "Grandfathered ingredient" authorized for use in Dietary supplements	Esters of propylene glycol
U.S.A.	FCC (Food Chemical Codex)	Propylene glycol mono- and diesters
U.S.A.	USFA US Food additives according to 21 CFR	21CFR §172.856 Propylene glycol mono-and diesters of Fats and Fatty acids
JAPAN	JSFA (Japanese Standards of Food Additives)	Propylene glycol esters of Fatty acids

HANDBOOK of FOOD ADDITIVES

Denomination	Use	Safety
LAUROGLYCOL™ 90 Propylene glycol monolaurate	Food additive	FDA 21CFR §172.856; E 477; DMF no. 7818; worldwide food additive status

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

Liability clause

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